

Mobile Mapping for OpenStreetMap

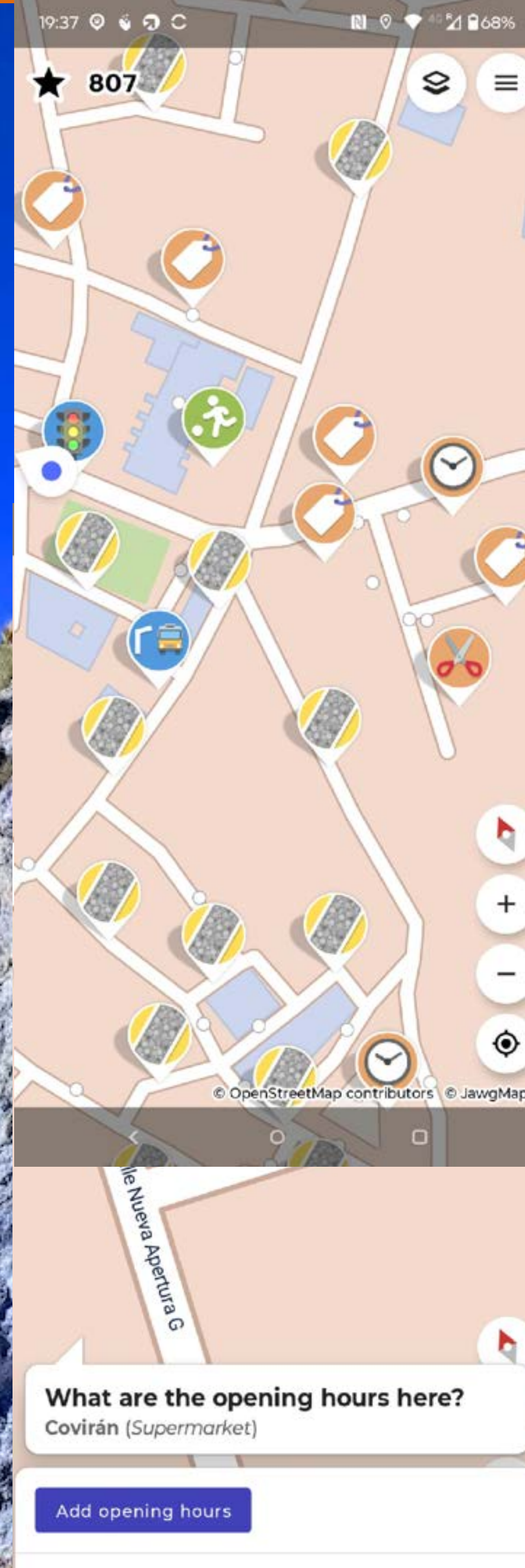
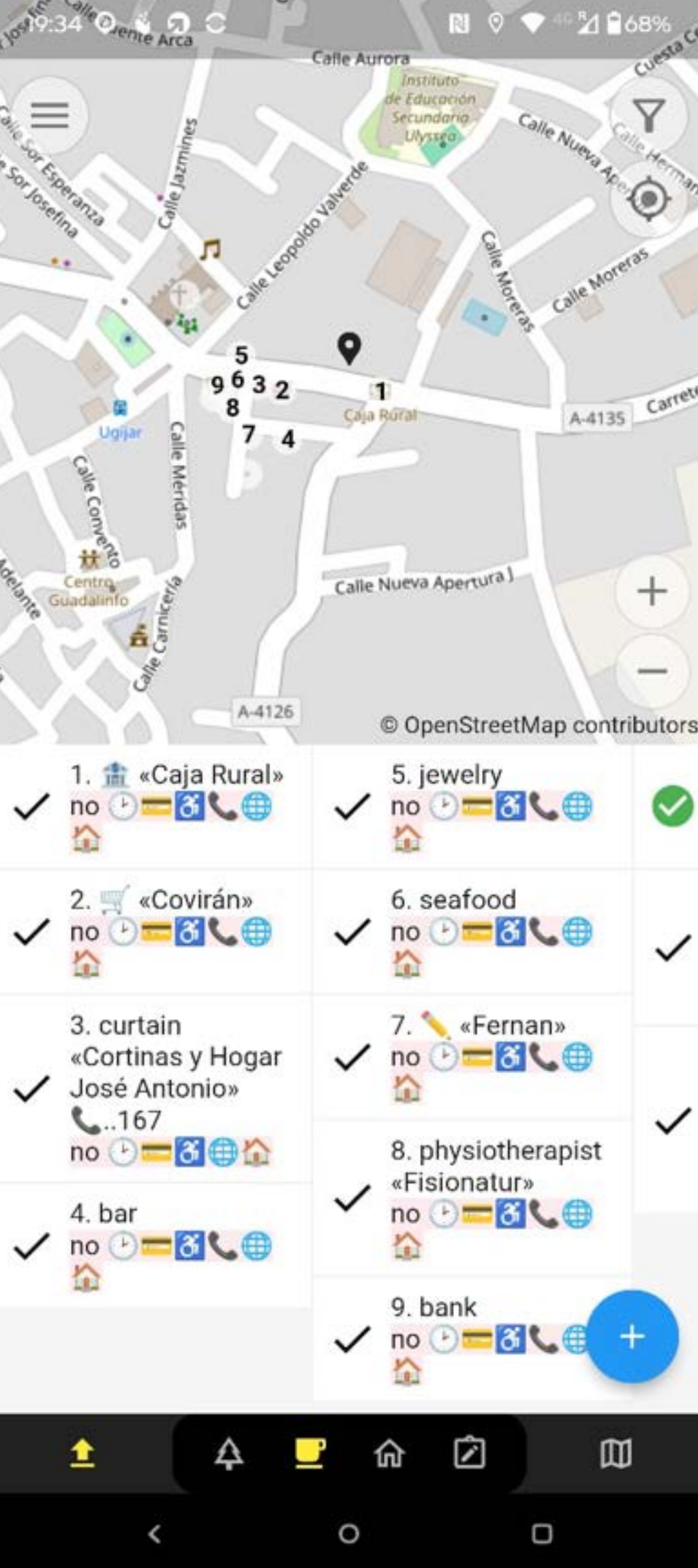


Taller - Workshop - (English)

*Learn how to map on OpenStreetMap, the free map of the world, the "Wikipedia of maps". Roads, paths, bars, shops,...
By: Just van den Broecke (OSM-NL)*

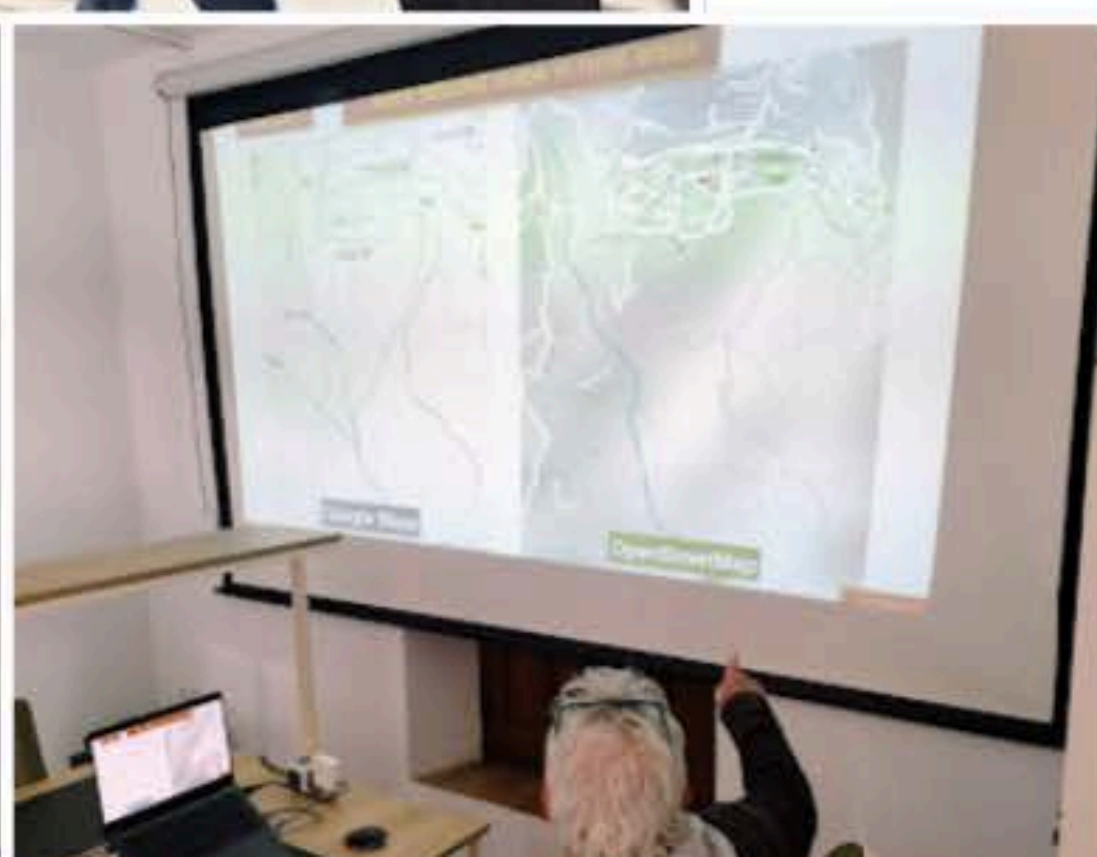
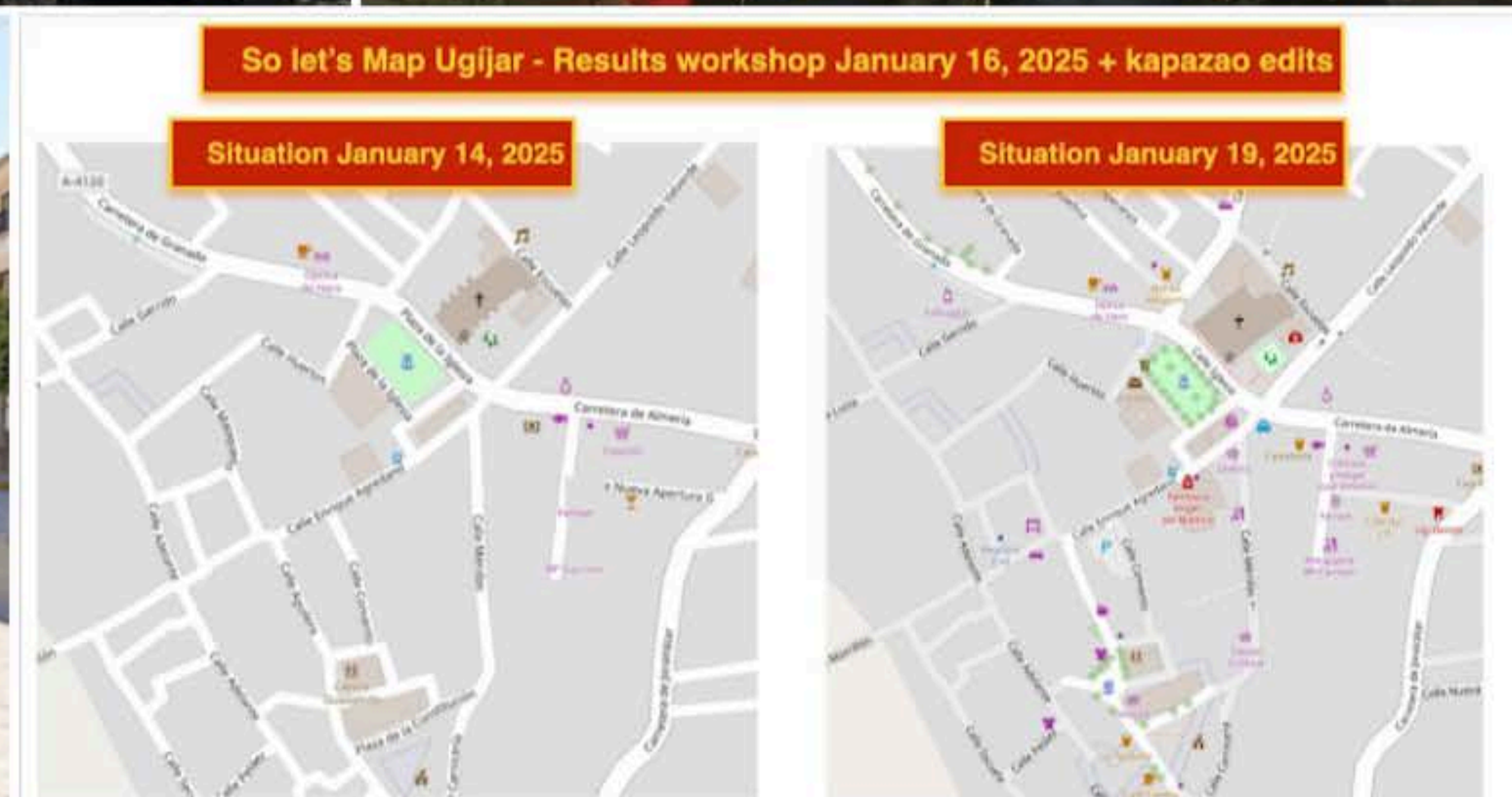
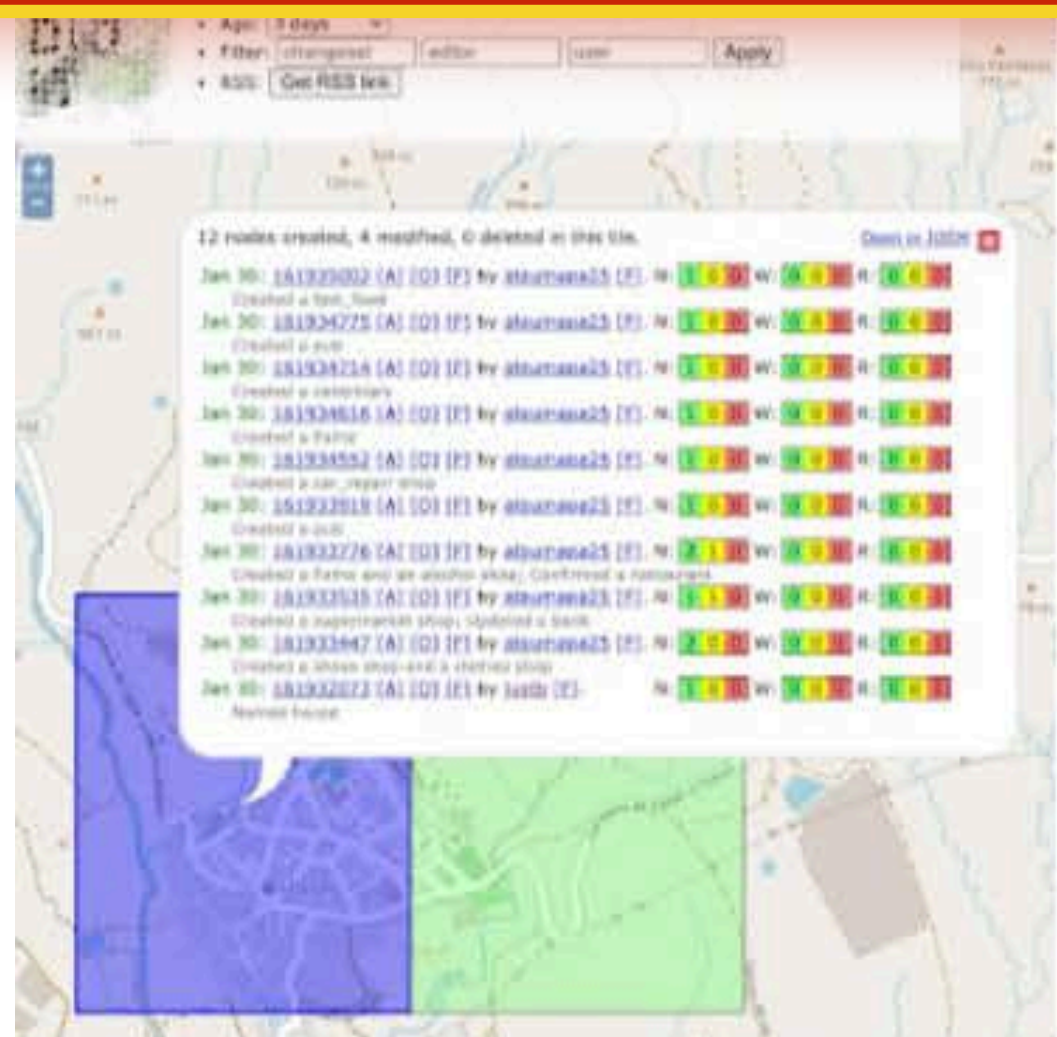
workshop docs
alpumapa.xyz

V2





Workshops in Spain



Agenda

- Introduction (presentation slides)
- Workshop via alpumapa.xyz
 - create account on openstreetmap.org
 - learn, install and configure apps
 - go on the streets to map! OR remain inside (armchair mapping)
BE CAREFUL WHILE LOOKING AT PHONE AND WALKING!
 - return here for looking at results & evaluation

OSM Website: openstreetmap.org

Just van den Broecke
Open Source
Geospatial Professional
www.justobjects.nl

*Soy
hollandés*

**About
Me**

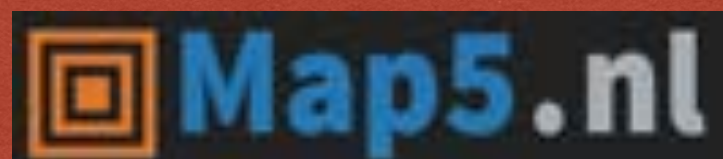
Founder



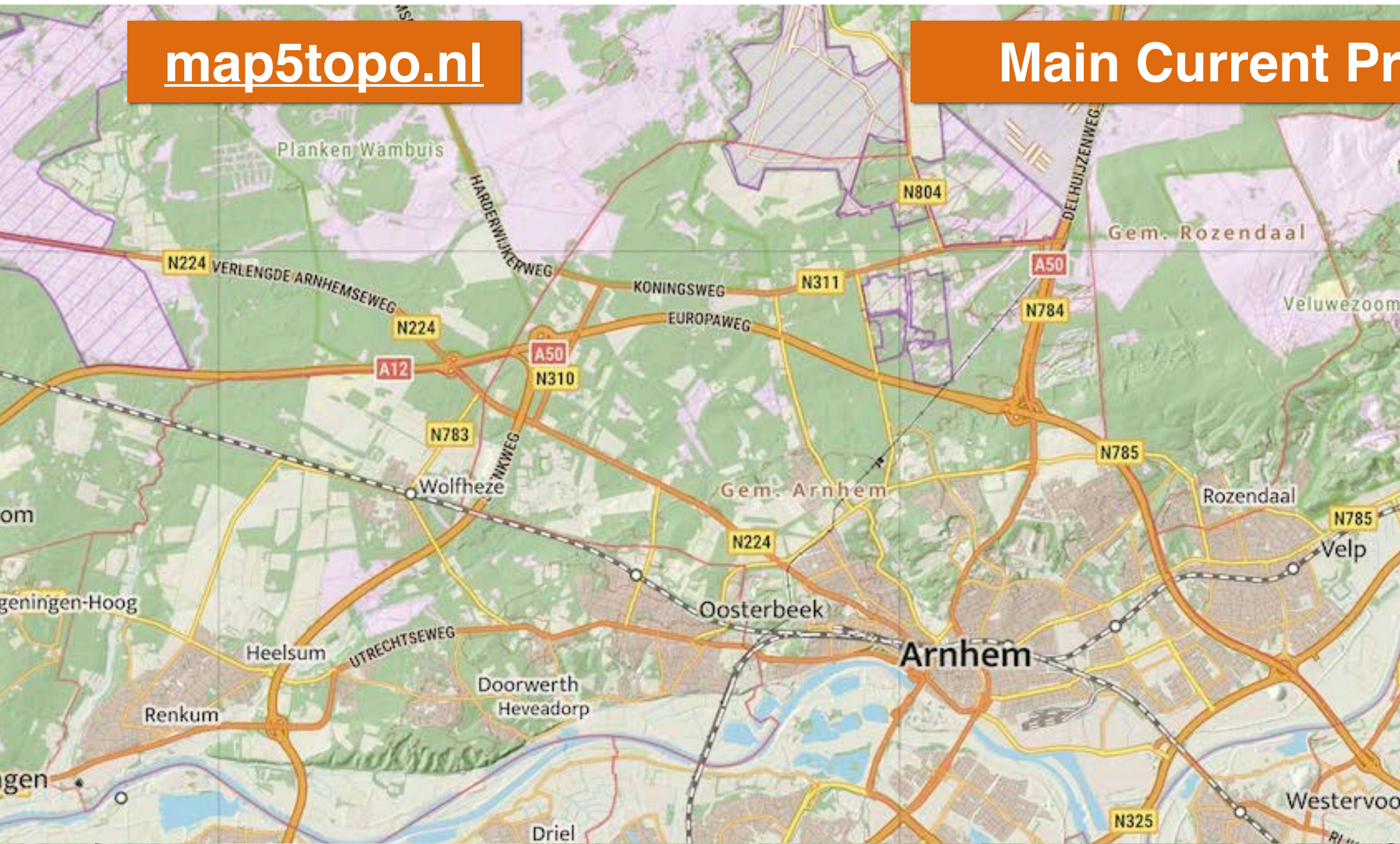
FOSS Projects



Cloud Services



OpenStreetMap Mapper
openstreetmap.org/user/justb



What is OpenStreetMap?

- **Definition:**
A free, open-source, and collaborative mapping platform where anyone can contribute, edit, and use map data.
- **Mission:**
To create and provide free geographic data to everyone.
- **Fun Fact:**
Often referred to as "the Wikipedia of maps."

OSM Website: openstreetmap.org

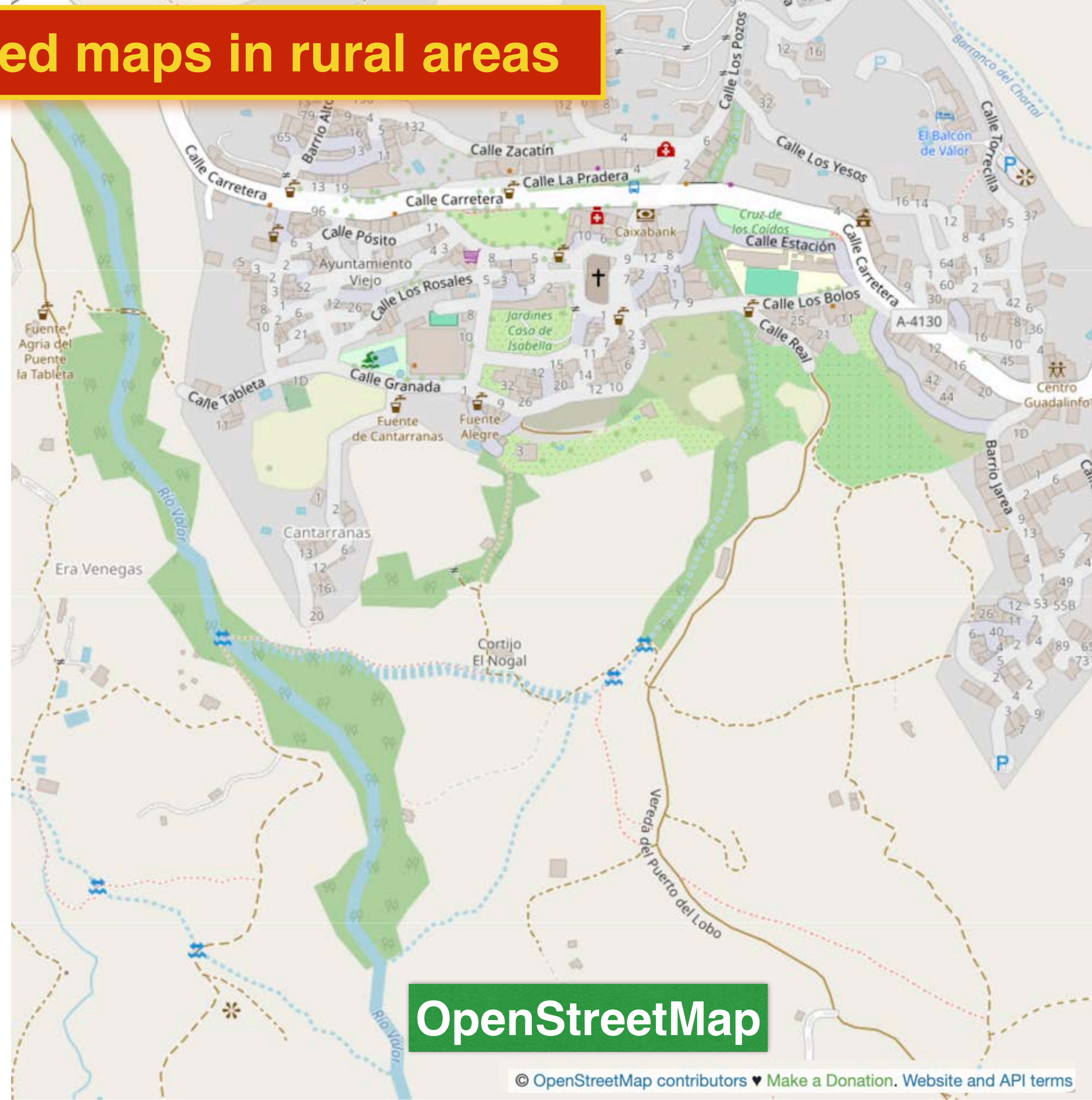
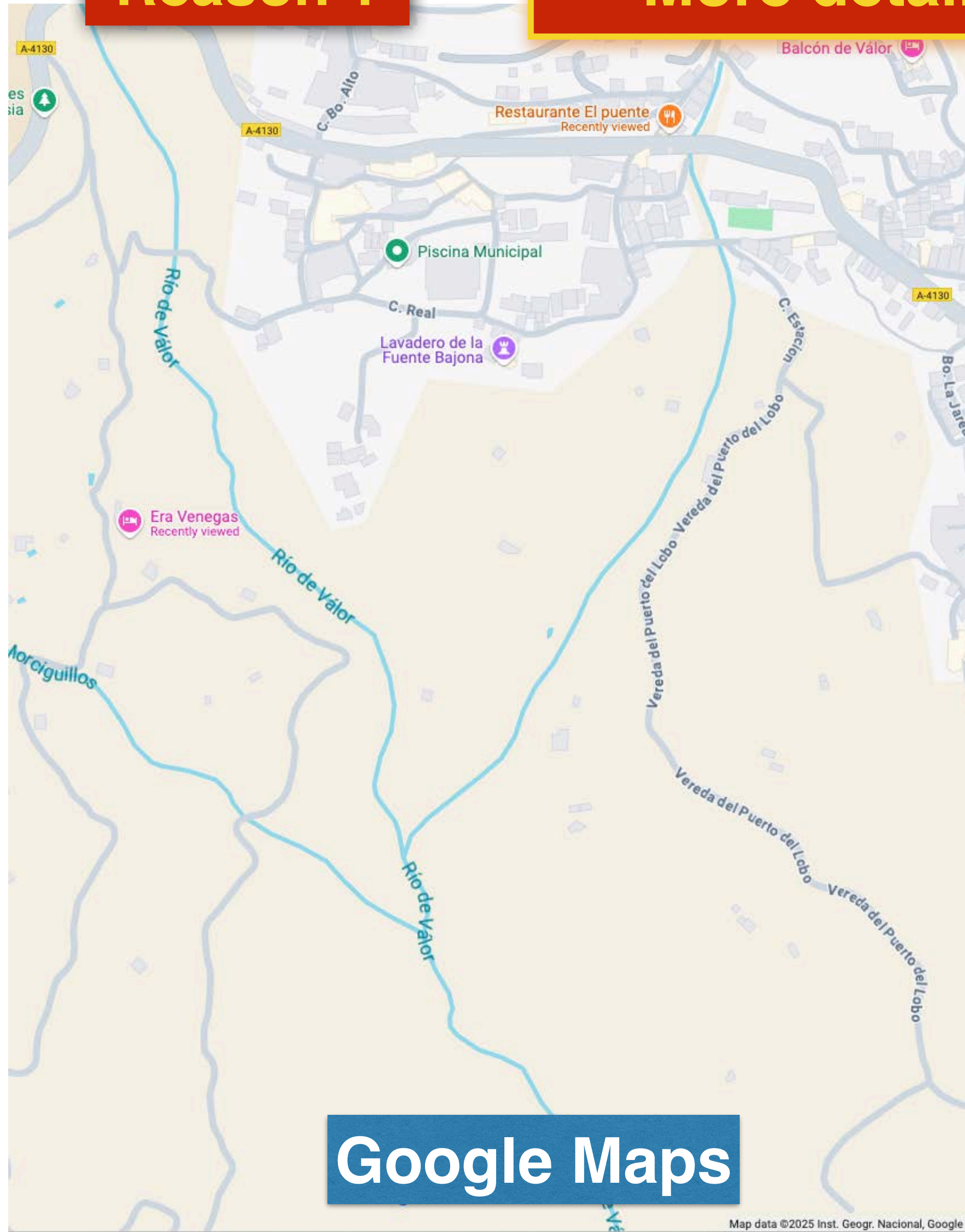


Why? Por que?

Three Reasons
Tres Razones

Reason 1

More detailed maps in rural areas

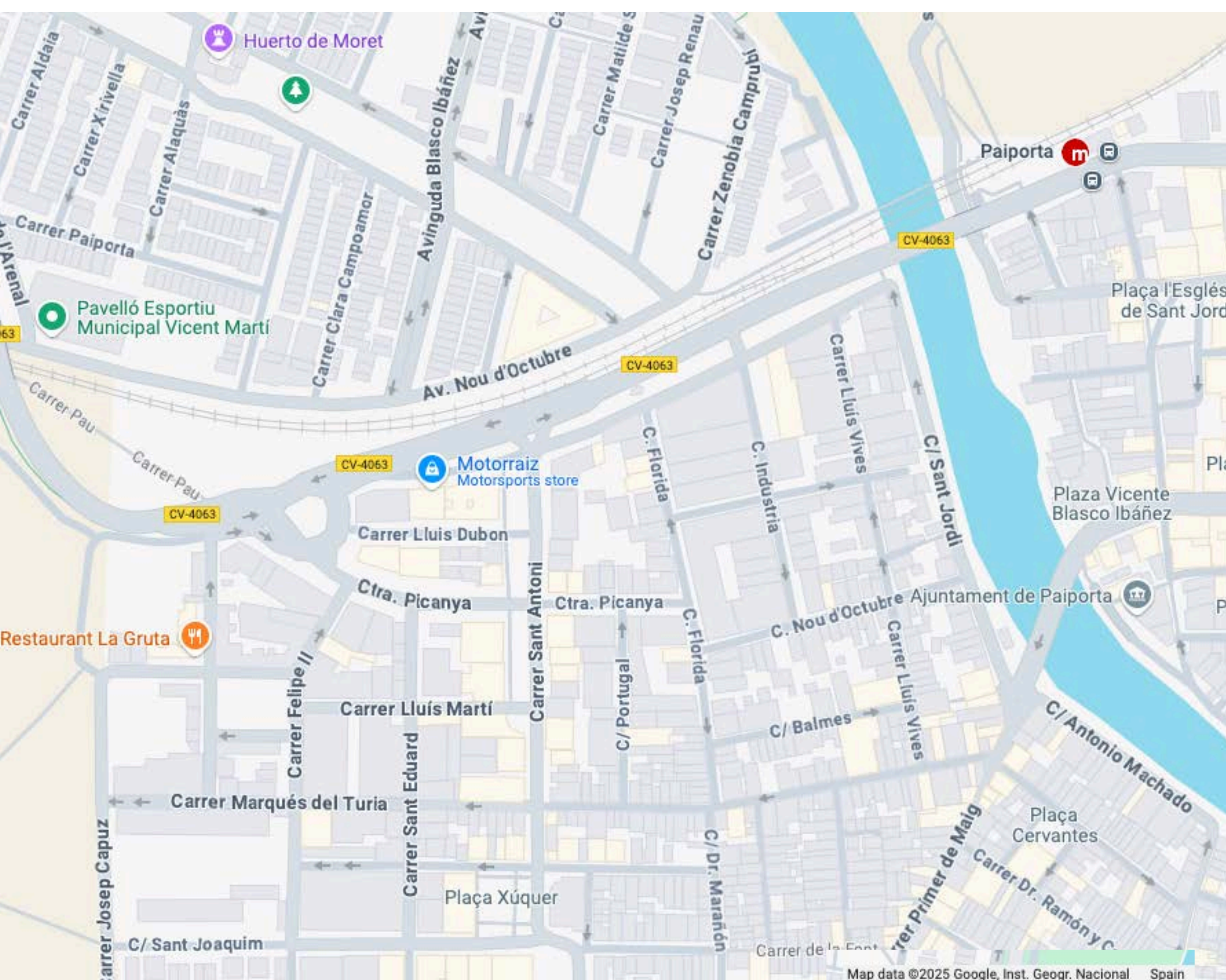


OpenStreetMap

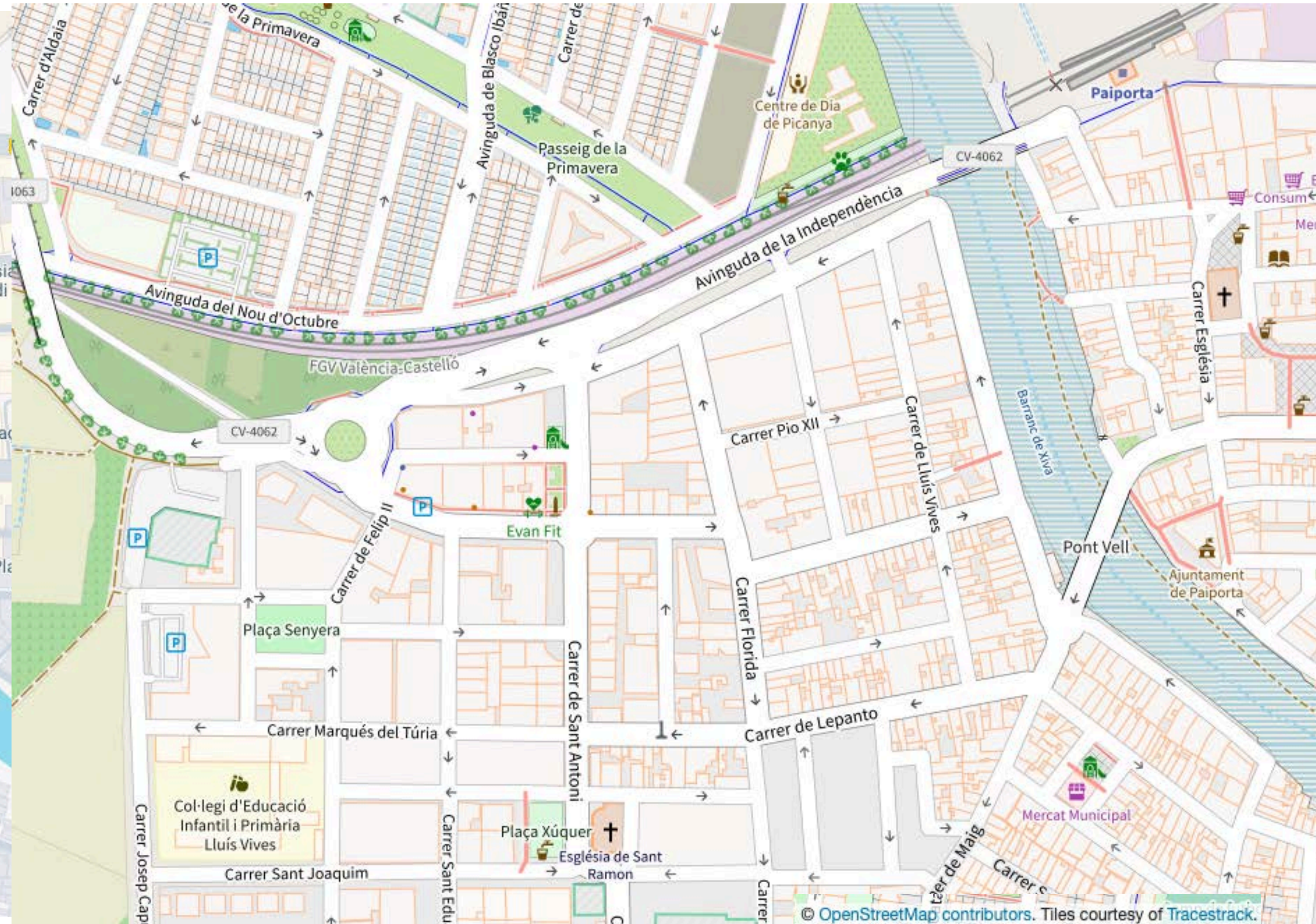
© OpenStreetMap contributors ♥ Make a Donation. Website and API terms

Reason 1

More detailed maps in general



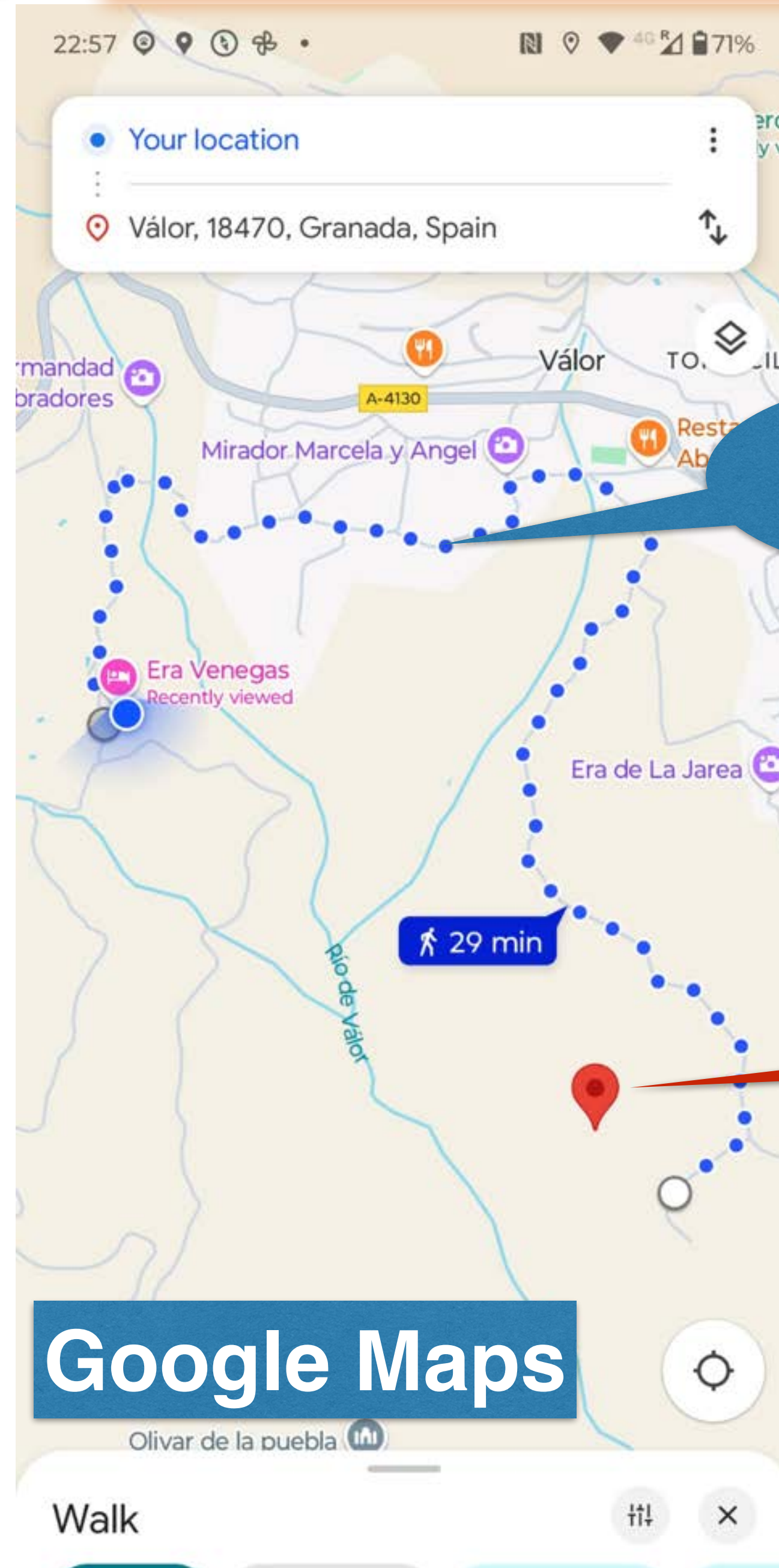
Google Maps



OpenStreetMap

Reason 1

Pedestrian Routing - Google Maps vs CoMaps (OSM)



Routing over
car roads: 29 min

"Take me to
G&T"



Reason 2

Disaster Response and Humanitarian Mapping

Mapping our world together

[HOT](#) Humanitarian OpenStreetMap Team

[WHAT WE DO](#) [OUR WORK](#) [TOOLS & DATA](#) [NEWS](#) [OPEN MAPPING HUBS](#) [COMMUNITY & ORGANIZATION](#) [PARTNER WITH US](#) [GET INVOLVED](#) [DONATE](#)

HOT IS AN INTERNATIONAL TEAM DEDICATED TO HUMANITARIAN ACTION AND COMMUNITY DEVELOPMENT THROUGH OPEN MAPPING.

[LEARN WHAT WE DO](#) [START MAPPING >](#)

491K
Total Mappers^①

260M
Total Map Edits^①

156.4M
Buildings Mapped^①

3.4M
Roads Mapped (KM)^①

[alpumapa.xyz](#)

Reason 2

Disaster Response and Humanitarian Mapping

Humanitarian OpenStreetMap Team
23,363 followers
2mo • 

#OSM Community, we urgently need your help to map roads and pathways in #Valencia and other affected municipalities due to severe flooding from #DANA. Your contributions can make a huge difference in the response efforts! Let's map: <http://bit.ly/SPNFloods24>

URGENT ACTIVATION



We require your **urgent support** mapping.

CAMPAIGN bit.ly/SPNFloods24

Spain Floods 2024 | tasks.hotosm.org

Mapping our world together

Humanitarian OpenStreetMap Team Website

Tasking Manager EXPLORE PROJECTS MY CONTRIBUTIONS LEARN ABOUT SUPPORT

#17948 | OSM España HIGH

SPAIN FLOODS 2024: VALENCIA ROADS AND PATHS (OTHER AFFECTED MUNICIPALITIES)

Spain Floods 2024 · Spain · disaster response

On 29 October 2024, torrential rain caused by an isolated low-pressure area at high levels brought over a year's worth of precipitation to several areas in southeastern Spain, including the Valencian Community. The resulting floodwaters caused the deaths of at least 205 people and substantial property damage. It is one of the worst natural disasters in the history of Spain by the number of confirmed deaths. See more in the [October 2024 Spain floods Wiki](#).

The OSM España community has identified it as a priority to complete the road network in the most affected region. Please contribute to this project to fill these data needs. See more in the [España/Humanitario/DANA 2024 Wiki](#).

Please map the priority area marked in red first. When that area is nearly finished, move to the rest

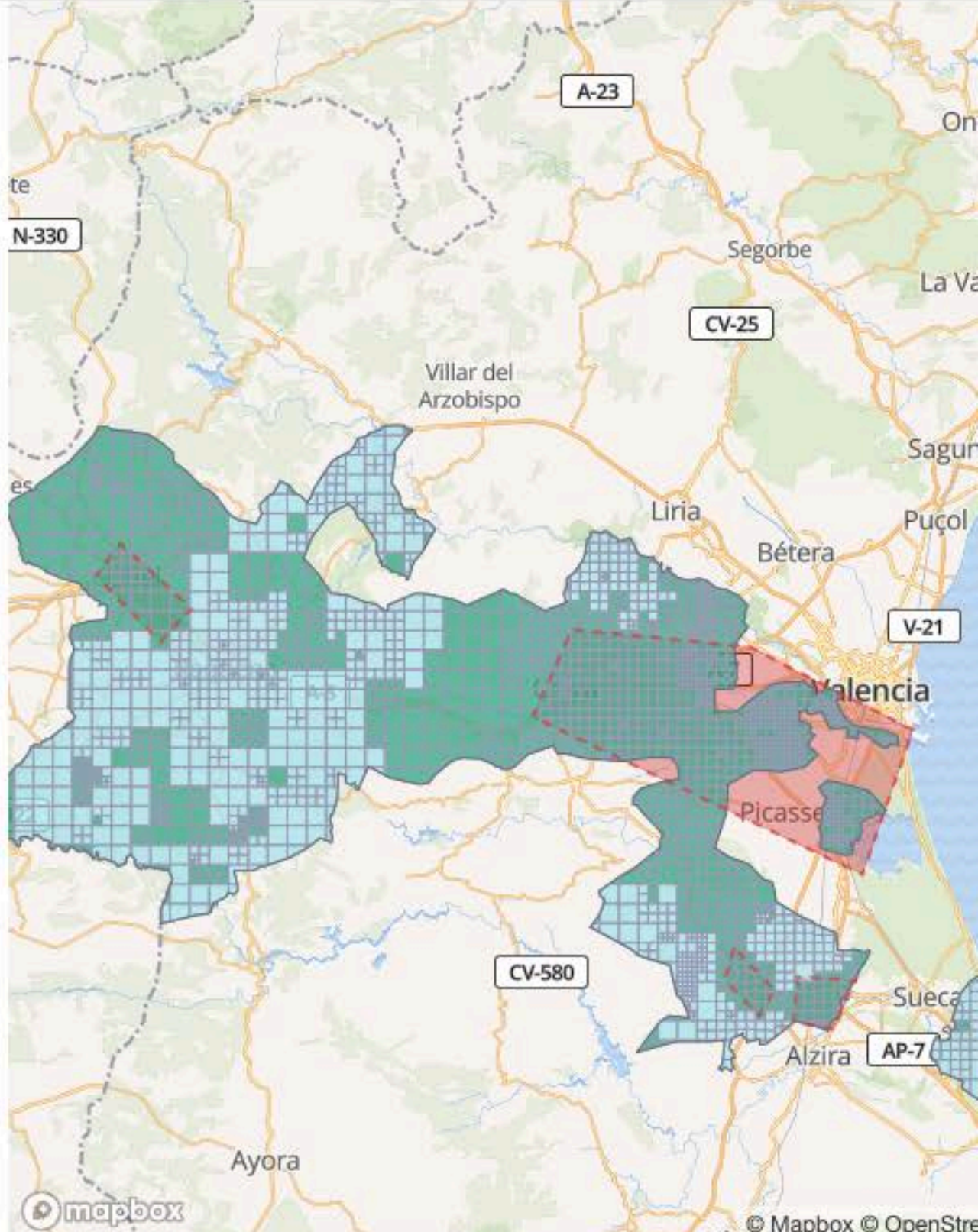
TYPES OF MAPPING     

IMAGERY **ESRI World Imagery**

694 contributors Last contribution 1 month ago

Moderate No due date specified

Overview · Description · Coordination · Teams & Permissions · Questions and comments · Contribution > Share Add to Favorites Contribute



**DANA Valencia (2024):
OSM played a crucial role in mapping flood-affected areas**

OSM España

516 members, 71 online

Pinned message #1

Bienvenida/o al grupo de mensajería instantánea de la Comunidad de OpenStreetMap en España NORMAS DEL GRUPO: 0- Este grupo es para hablar...

Translate to English

H ;)

Gente!! Activemos nuestro propio "HOT". A causa de la DANA varios puentes han caído y otras carreteras han desaparecido y van a tardar en volver. Especialmente zona de Valencia/Albacete

👍👤

La pasarela de Picanya ya la he quitado yo.

Si conocéis otra zona no dudéis de actuar sobre puentes y otros elementos que tardarán en volver. yo he añadido destroyed al bridge=yes quedando destroyed:bridge=yes

👍👤

He ido aquí ...pero no he visto nada. ¿No había una página con todas estas etiquetas?

https://wiki.openstreetmap.org/wiki/Disaster-specific_tags

Jordi M

Yo solo he encontrado información útil en la página de 'destroyed' (<https://wiki.openstreetmap.org/wiki/Key:destroyed>.)

Por cierto, 'destroyed:~' también debe de ponerse en la etiqueta highway, ¿no?

H ;)

sí

así lo he hecho, he dejado el nombre solo

la V-30 estará fuera de servicio algún día pero yo no la modificaría a no ser que se cayera algún puente

Tenemos trabajo. Id siguiendo las indicaciones de los vídeos

<https://www.levante-emv.com/videos/camp-turia/2024/10/29/puente-vilamarxant-completamente-agrietado-limite-110838782.html>

Levante-EMV

El puente de Vilamarxant, completamente agrietado y al límite de derrumbarse

fer

H ;)

Si conocéis otra zona no dudéis de actuar sobre puentes y otros elementos que tardarán en volver. yo h...

Yo hice esto mismo en 2 sitios por la dana del año pasado por el centro, luego lo arreglan y se quita el prefijo

Bienvenida/o al grupo de mensajería instantánea de la Comunidad de OpenStreetMap en España NORMAS DEL GRUPO: 0-

Translate to English

Jose Luis Infante admin

Así se ve desde el aire el destrozo que ha provocado la DANA en el by-pass | Las Provincias

<https://www.lasprovincias.es/comunitat/aire-destrozo-provocado-dana-bypass-20241031111747-nt.html>

Las Provincias

Así se ve desde el aire el destrozo que ha provocado la DANA en el by-pass | Las Provincias

El Ministro Óscar Puente describió este punto como «crítico» por su situación

Esto lo tenemos?

01 November 2024

Jordi M

Lo estoy mirando

Jose Luis Infante

Esto lo tenemos?

Editado: <https://www.openstreetmap.org/way/1317611449> y <https://www.openstreetmap.org/way/1317611451>

OpenStreetMap

Way: Autovía del Mediterráneo (1317611449)

OpenStreetMap is a map of the world, created by people like you and free to use under an open license.

Matrix Telegram Bridge

Nukeador: Hola, comentaba con Jordi M como podemos ayudar en la zona del DANA, si usando el <https://tareas.openstreetmap.es/explore> para coordinar mapeos

Jordi M

Bienvenido Nuke. Reenvío tus mensajes para que la gente sepa que ya te has presentado en el grupo valenciano de OSM:

Jordi M

Forwarded messages

Nuke 1/11/24

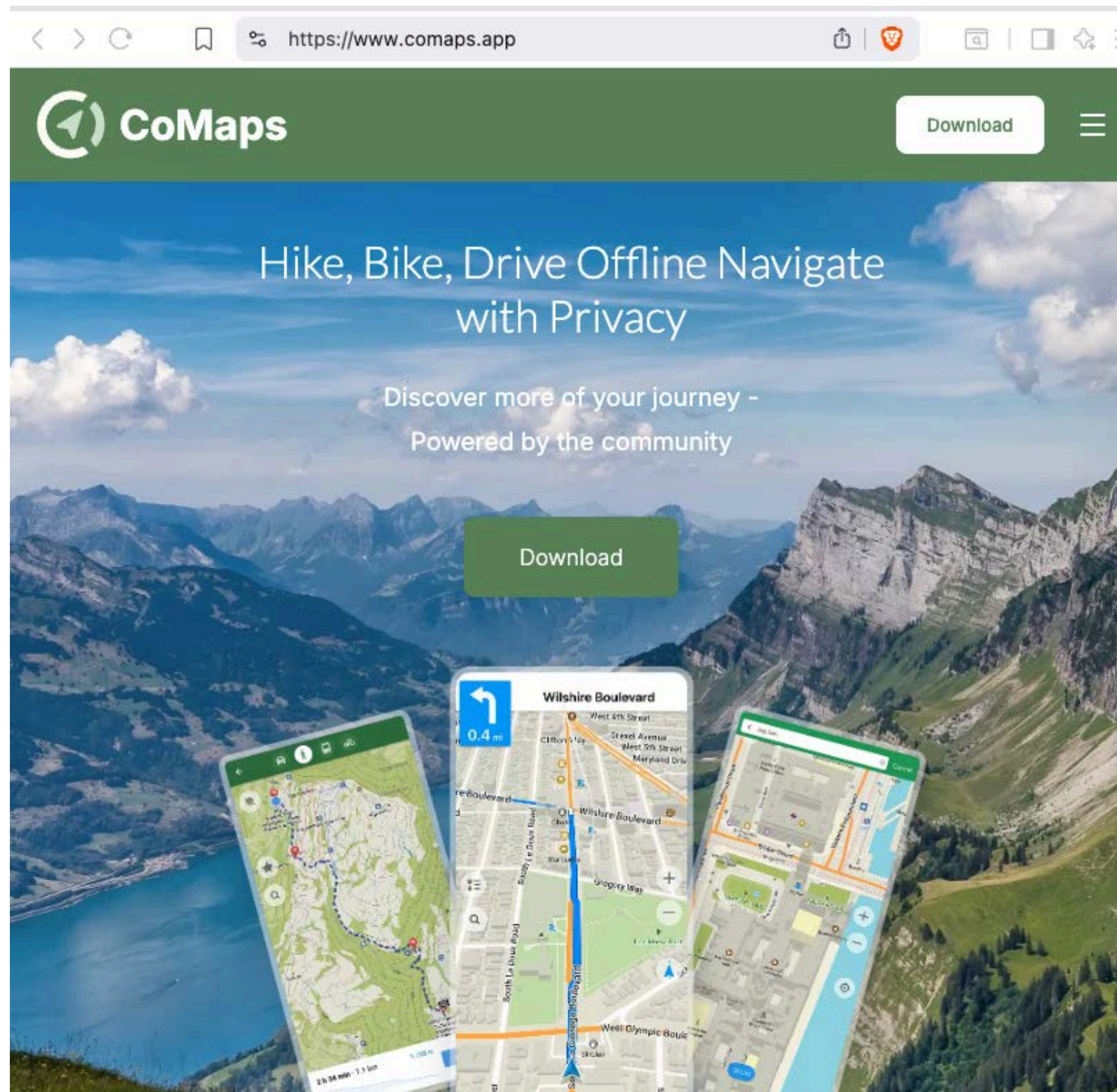
Hola, pertenecemos a la comunidad de datos abiertos BuscaBits en Valladolid, y nos preguntábamos cómo

alpumapa.xyz

- **People moving away from “USA Big Tech Apps”**
- **Switch to alternatives**
 - from Twitter/X/FaceBook to Mastodon
 - from WhatsApp to Signal
 - from Instagram to PixelFed
 - from YouTube to PeerTube
 - etc
- **OpenStreetMap fits in this trend**
 - from Google Maps to CoMaps, MagicEarth, OSMAnd..

Reason 3

OpenStreetMap Apps as Big Tech Alternatives



iOS

Android

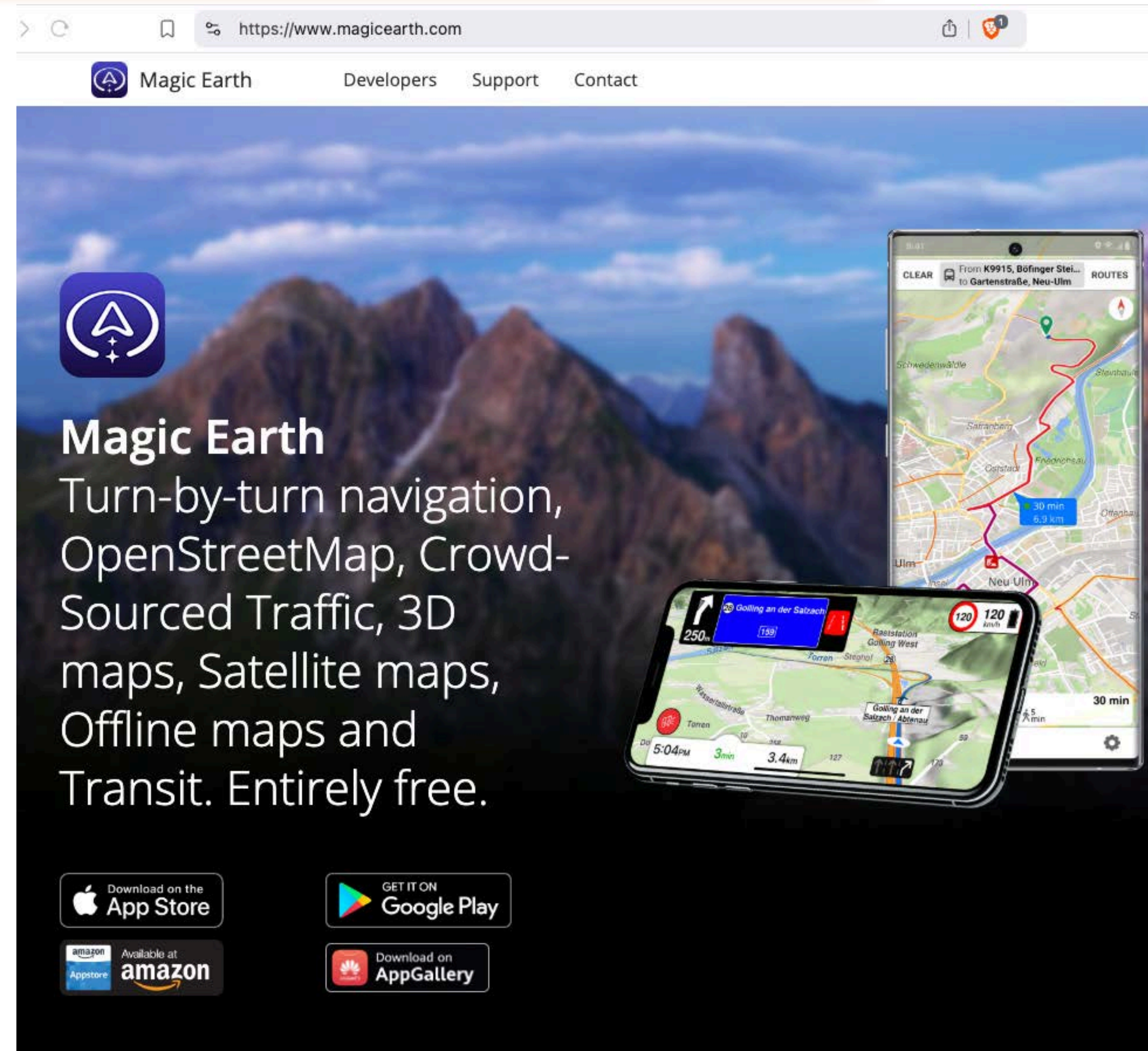
App Store

Google Play

F-Droid

Obtainium

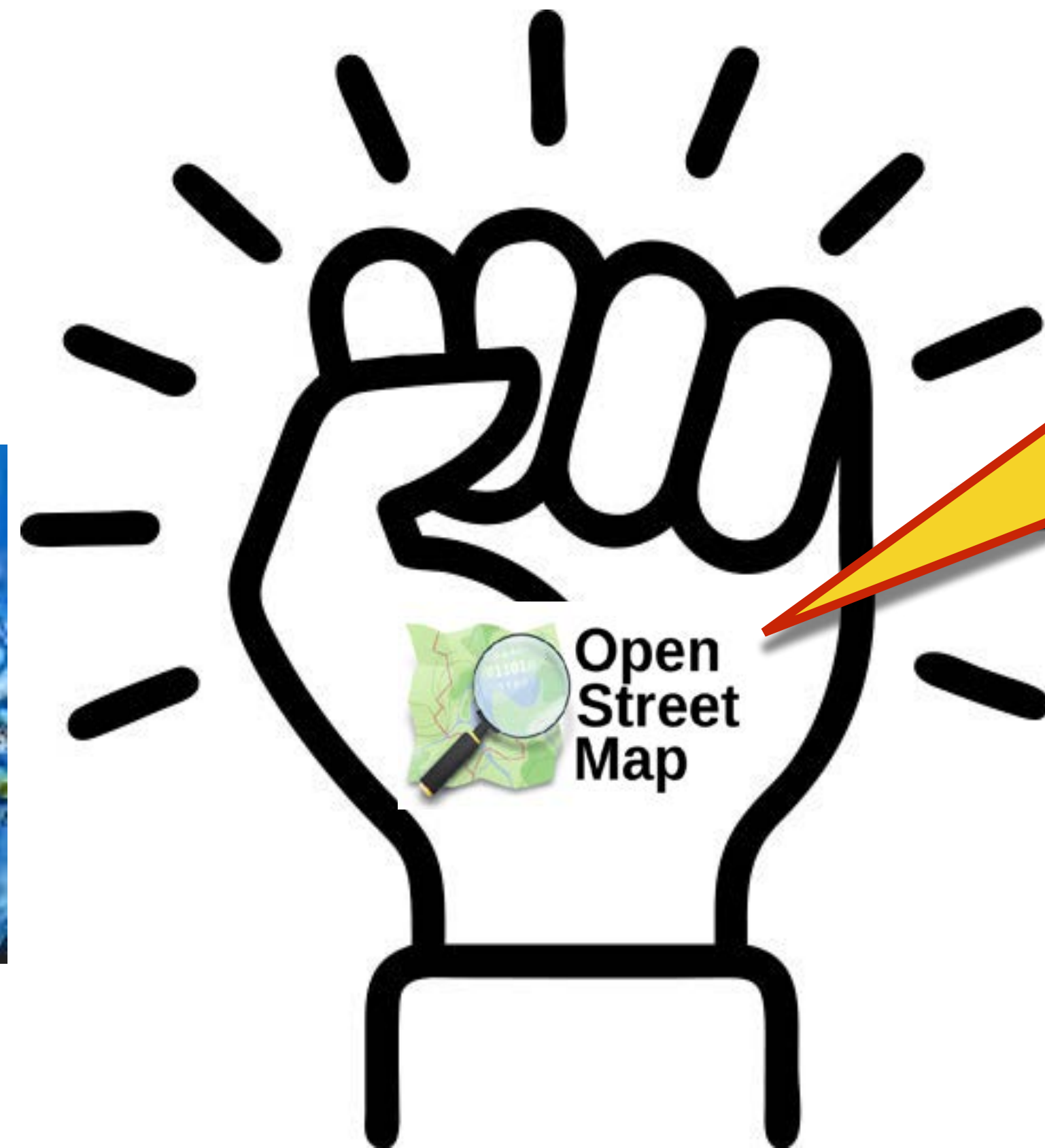
APK Download



So?

Three Reasons - One Word

*Never mind
the \$*@*% ,
Let's take Mapping in
our own hands*



Empowerment - Empoderamiento

OSM mapping for Válor - status July 6, 2025

Let's take
Mapping in
our own
hands

OSM mapping for Válor - status August 29, 2025

Let's take
Mapping in
our own
hands



Outdoor Apps Using OpenStreetMap

- **CoMaps:** offline maps with navigation for hiking and cycling
- **Komoot:** route planning for outdoor activities like hiking and biking
- **AllTrails:** provides trail maps and reviews for hiking and running
- **Wikiloc:** trail sharing and GPS navigation for outdoor enthusiasts
- **OsmAnd:** Offline maps with navigation and rich outdoor features

Editing and Contributing to OSM with Smartphone Apps

- **EveryDoor - [every-door.app](#)**
add or edit businesses, amenities, and more
- **StreetComplete - [streetcomplete.app](#)**
improve map by answering map-related questions
- **CoMaps - [comaps.app](#)**
also allows direct edits to OSM data
- **Many more like Vespucci, Go Map, MapComplete.. ,,**

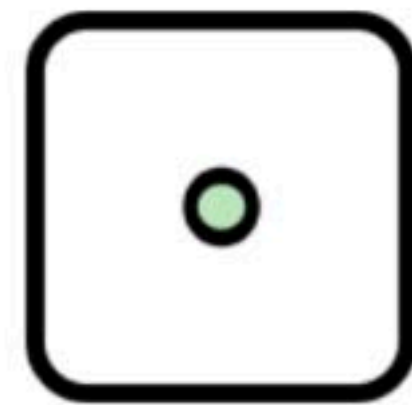
OpenStreetMap Inner Workings

- **IT IS ALL ABOUT DATA!**
could have called project *OpenGeoData*
- **OSM DATA IS IN ONE BIG DATABASE**
maps like openstreetmap.org made with this data
- **ECOSYSTEM OF “APPS” TO EDIT OSM DATA**
like Id, JOSM, EveryDoor, StreetComplete etc

Diving Deeper - How is OSM Data Structured?

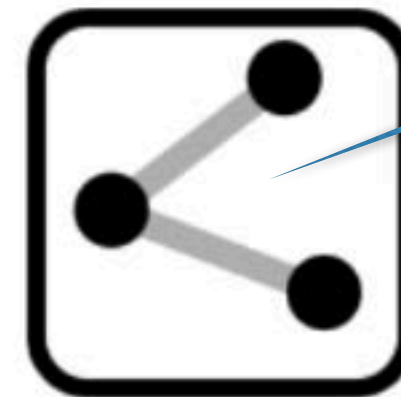
Also Closed
Ways: Areas like
Orchards,
Buildings,..

1. Core Elements:



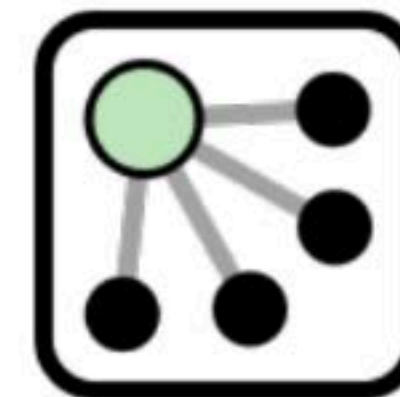
Node

“Points”



Way

“Lines”



Relation

“Combinations”

**Church,
Restaurant,
Bench etc**

**Roads, Paths,
Tracks, etc**

**Busline,
Hiking Route**

Examples:

wiki.openstreetmap.org/wiki/Beginners_Guide_1.3

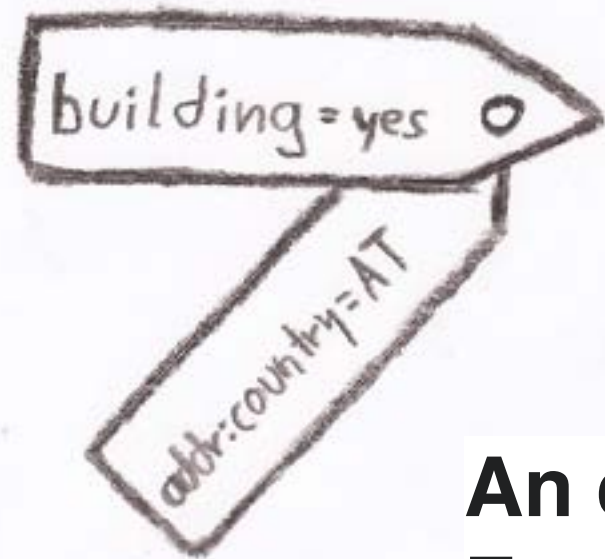
Nice Overview: jakobmikschi.eu/post/openstreetmap_overview/

Diving Deeper - How is OSM Data Structured?

2. Tagging

The Elements Nodes, Ways, Relations have “Tags”

Tags are textual key-value pairs that describe properties of an Element



An element can have **tags**. A tag is a **key=value** pair describing what the element is. For instance, mapping a mobile phone store can be done by creating a **node**, and adding the following tags: **shop=mobile_phone**, **name=John Smith's phone centre**.

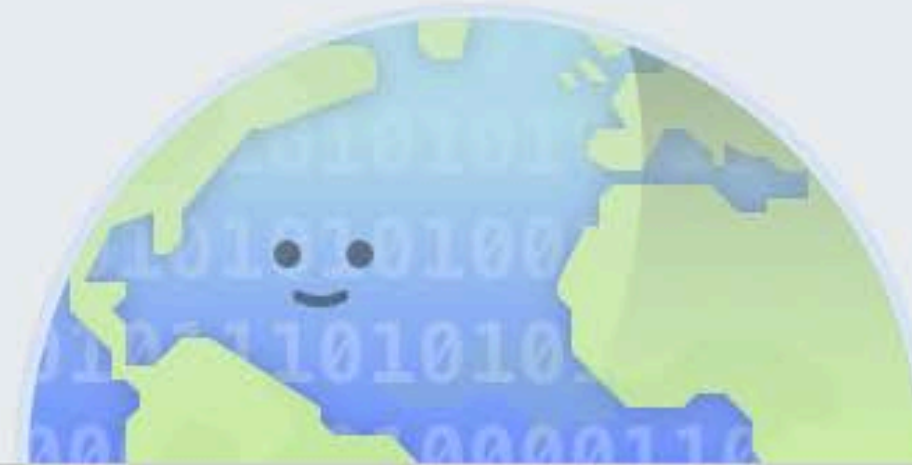
Here are some more examples:

- A parking lot can be a **closed way** tagged with **amenity=parking**
- A building can be a **closed way** tagged with **building=yes**
- A residential road can be a **way** tagged with **highway=residential**.
- A McDonald's branch can be a **node** tagged with **amenity=fast_food**, **brand=McDonald's**, **name=McDonald's**

wiki.openstreetmap.org/wiki/Beginners_Guide_1.3

Nice Overview: jakobmikschi.eu/post/openstreetmap_overview/

Signup at openstreetmap.org/user/new

[Edit](#)[History](#) [Export](#) [GPS Traces](#) [User Diaries](#) [Communities](#) [Copyright](#) [Help](#) [About](#)[Log In](#)[Sign Up](#)[Log in](#)[Sign up](#)

Free and editable. Unlike other maps, OpenStreetMap is completely created by people like you, and it's free for anyone to fix, update, download and use.

Sign up to get started contributing.

Email

Your address is not displayed publicly, see our [privacy policy](#) for more information.

Display Name

Your publicly displayed username. You can change this later in the preferences.

Password

Confirm Password

By signing up, you agree to our [Terms of Use](#), [privacy policy](#) and [contributor terms](#).

☐ I consider my contributions to be in the [public domain](#).

[Sign Up](#)

or sign up with a third party



Navigating the openstreetmap.org website

The image shows a screenshot of the OpenStreetMap website interface. The browser address bar displays the URL: <https://www.openstreetmap.org/#map=17/36.962893/-3.051941>. The top navigation bar includes links for History, Export, GPS Traces, User Diaries, Communities, Copyright, Help, and About. The user profile 'justb' is visible in the top right corner. The main map area shows a street grid in a city, with various landmarks and roads labeled. Several blue callout bubbles with white text and arrows point to specific features on the website:

- Edit the Map (OSM Data)**: Points to the 'Edit' button in the top left corner.
- Your account**: Points to the user profile 'justb' in the top right corner.
- Zoom in/out**: Points to the zoom controls (plus and minus buttons) on the right side of the map.
- Change background layer + overlays**: Points to the map style controls (satellite, map, and other overlays) on the right side of the map.
- Legend**: Points to the legend icon (a square with a question mark) on the right side of the map.
- Add a note to the map**: Points to the 'Add note' icon (a speech bubble) on the right side of the map.
- Show OSM data at click location**: Points to the 'Show OSM data' icon (a magnifying glass over a location pin) on the right side of the map.
- Search and jump to place**: Points to the search bar in the top left corner.
- Create a route**: Points to the 'Where is this?' button in the top left corner.

The bottom left corner of the map shows a scale bar for 50 m and 300 ft. The bottom right corner of the map shows the copyright notice: © OpenStreetMap contributors. Make a Donation. Website and API terms.

ON TO THE WORKSHOP

GO TO alpumapa.xyz

Thanks! Questions?

Home: justobjects.nl

Mastodon: mapstodon.space/@justb4

LinkedIn: linkedin.com/in/justb4/

OSM Workshop: alpumapa.xyz

CoMaps Workshop

- **Installation**
- **Download map area**
- **Explore the map**
- **Explore settings**
- **Navigation (foot, car)**
- **Working with GPS tracks**
 - **Sharing**
 - **Recording**
 - **Creating (gpsvisualizer.com/draw)**
- **Advanced: Edit OpenStreetMap**

CoMaps Workshop - Docs

<https://alpumapa.xyz/apps/comaps/>

<https://alpumapa.xyz/es/apps/comaps/>



CoMaps Workshop - WIFI

WIFI Name: MAPAS_TAPAS

Password:

CoMaps Installation

<https://www.comaps.app/download/>



- **Choose Store: App Store (iPhone) - Play Store (Android)**
- **Install**
- **On first open will ask to download map around current position**

Explore the Map

<https://www.comaps.app/support/>



- **Panning and Zooming**
- **Map Styles and Layers - “Outdoors” and “Contour Lines” BEST!**
- **Map orientation: Fixed North Up or Floating**

Navigation

- Find your favorite place - Tap and Select “Route To”
- Alternate mode of transport (foot, car, ...)
- Search place (looking glass), then “Route To:
Hint, enter: “Ginny & Terry”
- Try “Route From” with “Route To”, and Saving route
- We will practice outside later



Working With GPS Tracks

- “GPX File” universal file format tracks and routes
- Download from any place: Wikiloc, AllTrails etc
- Share via WhatsApp, Signal, or any Cloud storage
- We will practice outside later

Tip: <https://infraestructuras.fadmes.es/senderos>



Senderos Andalucía



SENDEROS

ESCALADA

VÍAS FERRATAS

BARRANCOS

ROCÓDROMOS

REFUGIOS

PR-A 480 VÁLOR - CUESTA VIÑAS - YEGEN - LA SALUD

Origen: Localidad de Válor

Destino: Localidad de Válor

Provincia: Granada

Municipio: Alpujarra de la Sierra, Válor

Datos del sendero

Estado del sendero: Homologado
Itinerario: Circular
Longitud: 12.84 km
Tiempo de ida: 3 horas y 40 minutos
Altitud máxima: 1370 m
Altitud mínima: 894 m
Desnivel positivo: 870 m
Desnivel negativo: 870 m
Época idónea: Todas excepto verano

Dificultad MIDE:

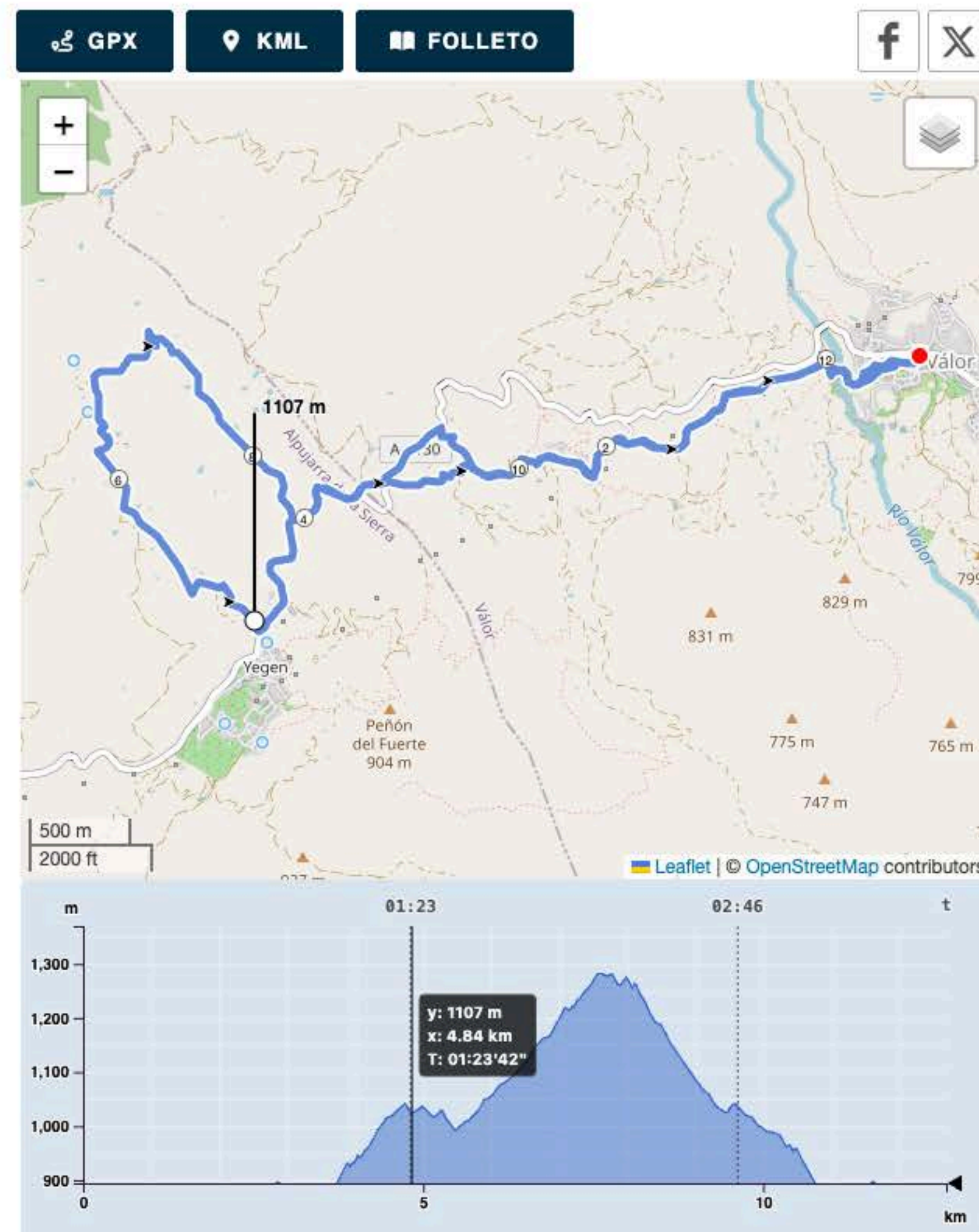


1 Valor mínimo, 5 valor máximo

Dificultad SIDIF:



Etiquetas experienciales:

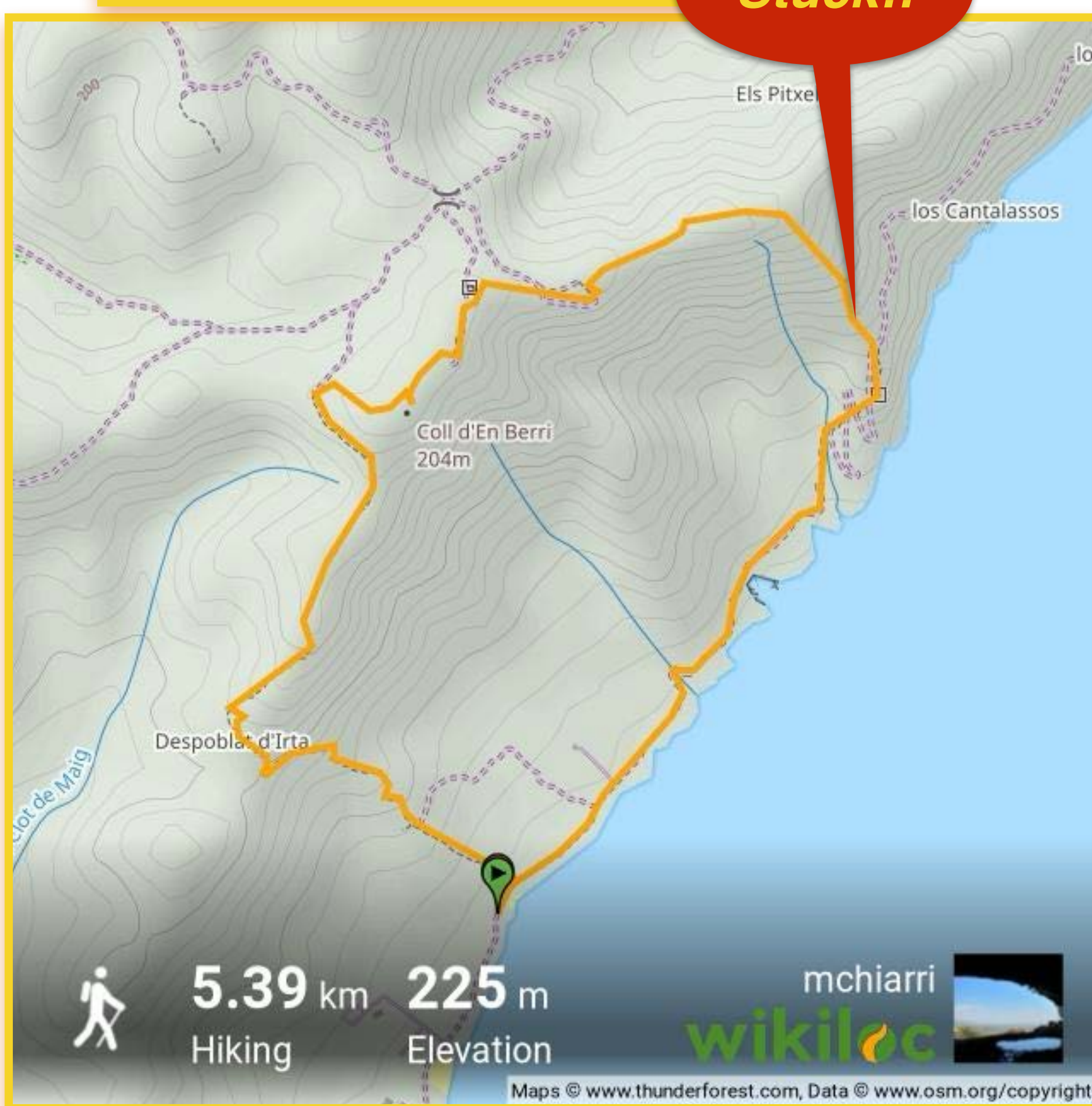


PR-A 480 VÁLOR LA SALUD

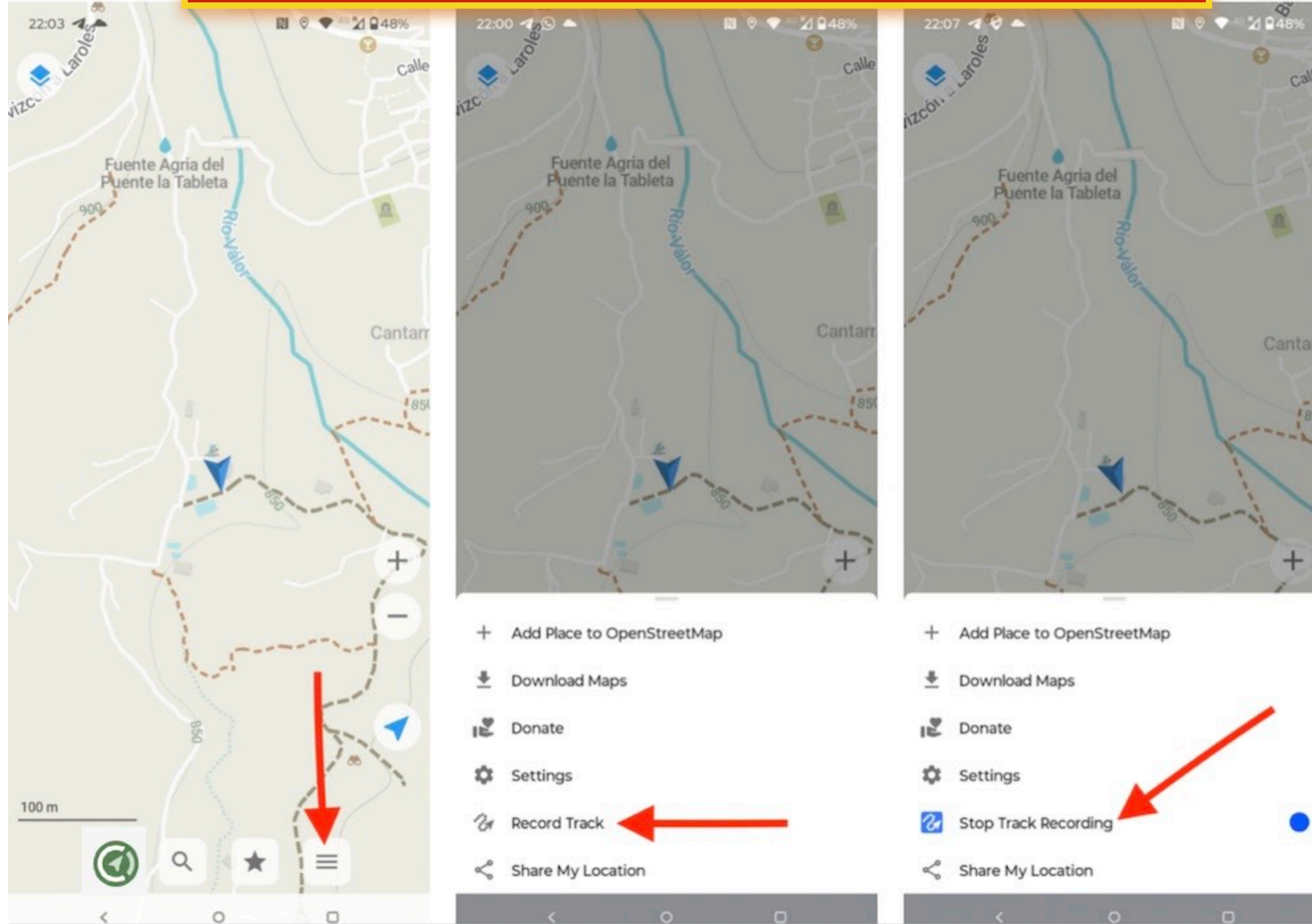


Beware of rutas malas!

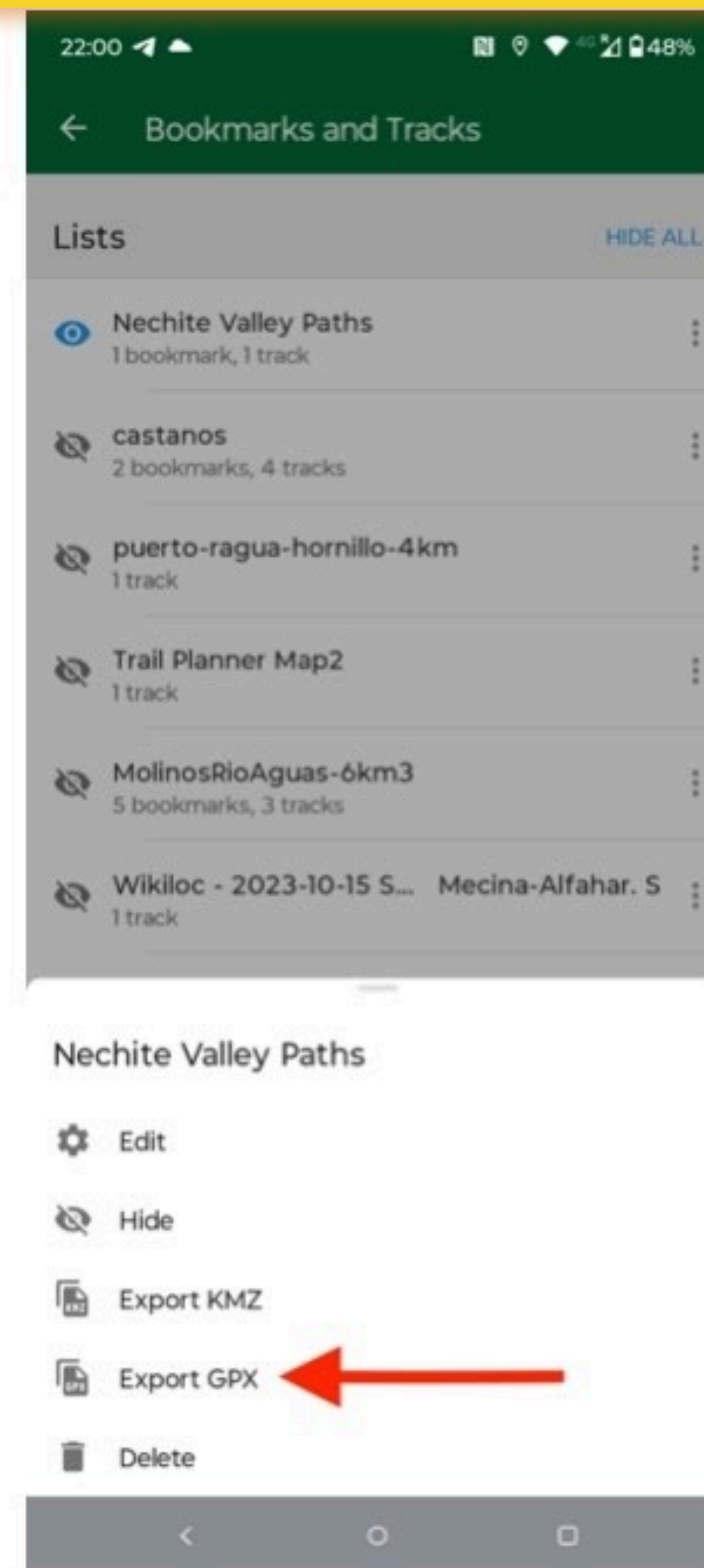
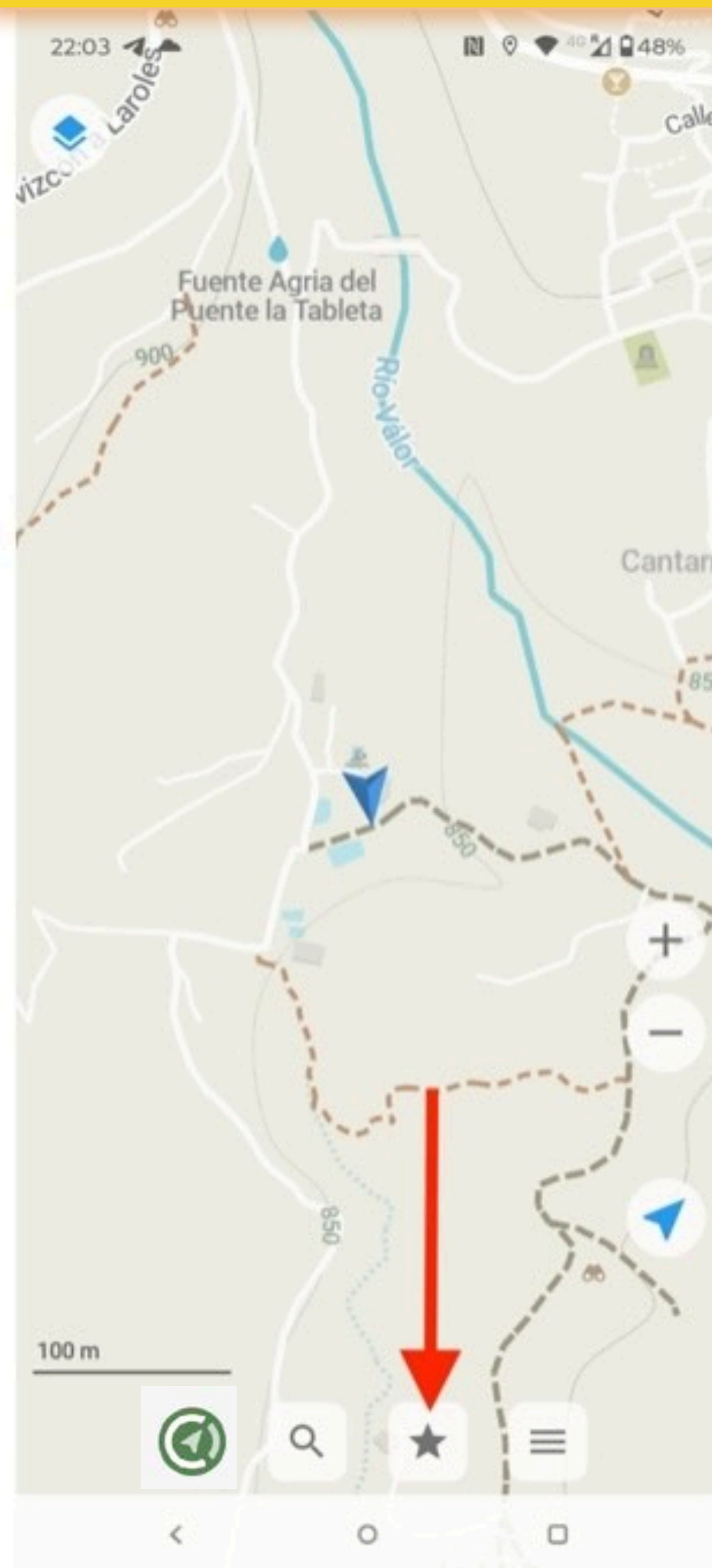
Stuck!!



CoMaps - GPS Track Recording



CoMaps - GPX Track Export



Creating your own route - [gpsvisualizer.com](https://www.gpsvisualizer.com)

<https://www.gpsvisualizer.com/draw/>

GPSVisualizer
Freehand Drawing Utility ("The Sandbox")
[\[Click here for the Google Maps version\]](#)

Use the toolbar on the right to add waypoint markers and tracks, then click one of the "save data" buttons to download as plain text, GPX, or Google Earth KML.

Center the map on a location:

OSM Nominatim found: **Válor, Comarca de la Alpujarra Granadina, Granada, Andalusia, 18470, Spain** (36.99617,-3.081) [precision: municipality]

Drawing tools:
pan wpt **trk** save data

Your map elements:
[track 5]

[Import a GPS file](#) [Import a URL](#)

Length: 0.485 km (0.301 mi)

Mouse: 36.99319,-3.08125
Center: 36.99431,-3.08172

Map created at [GPSVisualizer.com](https://www.gpsvisualizer.com)



GPSVisualizer Draw

Thanks! Questions?

Home: justobjects.nl

Mastodon: mapstodon.space/@justb4

LinkedIn: linkedin.com/in/justb4/

OSM Workshop: alpumapa.xyz